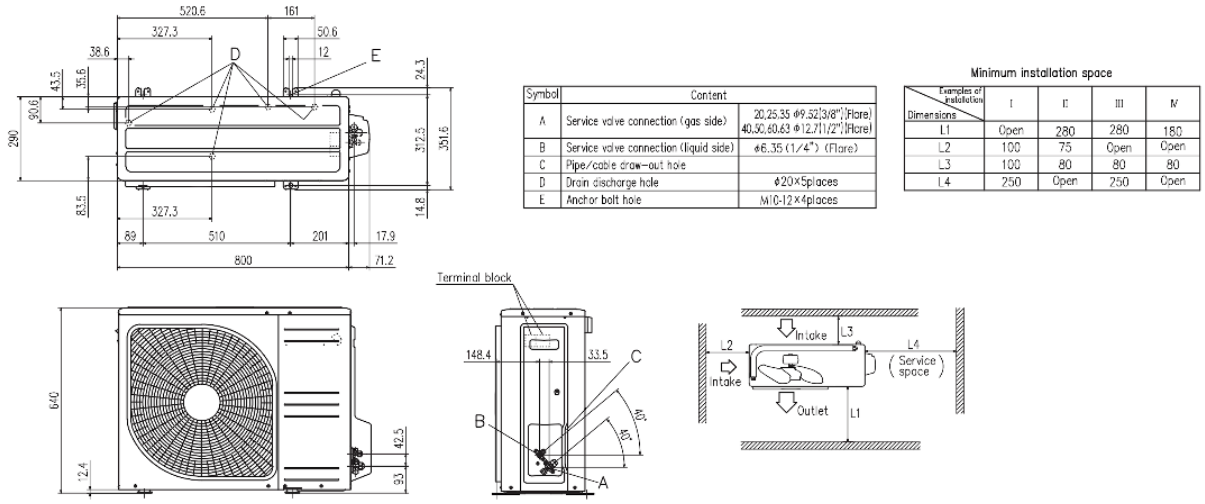


Figure 2.9 An engineering drawing of an external reverse cycle air conditioning unit. © Mitsubishi Heavy Industries.

2.4.2 3D geometric models

While 2D models convey layout logic, many digital twins require the depth and detail provided by 3D geometric models. These models form the visual and spatial backbone of digital twin applications, transforming abstract data into intuitive, interactive representations.

The core value of 3D models is that they:

1. Enable exploration of complex systems from multiple perspectives.
2. Show data within a spatial context, making it meaningful and actionable.
3. Support collaborative design before construction.

Applications require different 3D approaches, each offering distinct advantages: mesh-based models prioritize performance, parametric designs capture engineering intent, and point clouds preserve precise measurements.

BUILDING INFORMATION MODELS

A *building information model (BIM)* is a 3D model of a building or infrastructure asset that also contains rich information about the building's components, materials, and performance characteristics. BIM models